

```

26. # Update price list and recalculate MAs
27. prices.append(new_price)
28. prices.pop(0) # Remove oldest price
29.
30. short_ma = sum(prices[-short_period:]) / short_period
31. long_ma = sum(prices) / long_period
32.
33. # Check for trading signals
34. if short_ma > long_ma and position == "Flat":
35.     print("BUY signal generated!")
36.     position = "Long"
37. elif short_ma < long_ma and position == "Long":
38.     print("SELL signal generated!")
39.     position = "Flat"
40. else:
41.     print(f"Maintain {position} position")
42.

```

This simplified trend-following system:

1. Calculates short and long-term moving averages
2. Generates initial position based on their relationship
3. Updates the analysis as new prices arrive
4. Issues buy or sell signals when the moving averages cross

While basic, this structure mirrors how many professional trend-following systems operate, just with more sophisticated indicators and risk management.

Example 2: Portfolio Rebalancing Algorithm

1. This example shows how loops and conditionals can automate portfolio management:
2. # Portfolio holdings and target allocations